

Jaesik Park

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WORKING EXPERIENCES

Associate Professor

CSE and IPAI, Seoul National University (SNU)

Seoul, Republic of Korea

2025.09 – Present

- Principal Investigator of Visual and Geometric Intelligence Lab.
- Chief Professor of Student Affairs, IPAI (2024.09 – Present)
- Associate Director, Haedong Advanced Engineering Center (2025.05 – 2026.05)

Assistant Professor

CSE and IPAI, Seoul National University (SNU)

Seoul, Republic of Korea

2023.09 – 2025.08

Associate Professor

CSE and GSAI, Pohang University of Science and Technology (POSTECH)

Pohang, Republic of Korea

2022.09 – 2023.08

- Advised about 20 masters and Ph.D. students
- Selected as one of the POSTECH's Representative Research Achievements (2021)
- Received Outstanding Online Class Award (2020) and the Best EduTech Award (2021)

Assistant Professor

CSE and GSAI, Pohang University of Science and Technology (POSTECH)

Pohang, Republic of Korea

2019.04 – 2022.08

Staff Research Scientist

Intelligent Systems Lab, Intel (Manager: Dr. Vladlen Koltun)

Santa Clara, CA, USA

2015.12 – 2019.03

- Advised intern students from Stanford University, U.C. Berkeley, and Carnegie Mellon University
- **Co-creator of Open3D**: open-sourced 3D vision library built from scratch (**11.5+1.9k GitHub stars**)

EDUCATION

Ph.D. and M.S. in Electrical Eng.

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, Republic of Korea

2009.02 – 2011.02 & 2011.02 – 2015.08

- Ph.D. thesis: Image-based 3D Modeling via Constrained Optimization (Advisor: Prof. In So Kweon, Co-advisor: Dr. Yu-Wing Tai)
- Master's thesis: Upsampling Low-resolution Image using Heterogeneous High-resolution Image (Advisor: Prof. In So Kweon)

B.E. in Media Communication Eng.

*Hanyang University (*Summa cum laude*)*

Seoul, Republic of Korea

2005.03 – 2009.02

PUBLICATIONS

International

- [1] Cheolhong Min, Jaeyun Jung, Daeun Lee, Hyeonseong Jeon, Yu Su, Jonathan Tremblay, Chan Hee Song, and **Jaesik Park**
Why Far Looks Up: Probing Spatial Representation in Vision-Language Models
European Conf. on Computer Vision (**ECCV**), 2026
- [2] Kwonyoung Ryu, In-Jae Lee, Jonghyun Jin, Hyunjee Lee, Jongmin Lee, and **Jaesik Park**
Extending a Large View Synthesis Model for Multi-view Panoptic Segmentation
European Conf. on Computer Vision (**ECCV**), 2026
- [3] Daeun Lee, Jaeah Lee, Woosung Kim, Haebeom Jung, and **Jaesik Park**
Articulated Object Reconstruction from Rest-State Observation
European Conf. on Computer Vision (**ECCV**), 2026
- [4] Daneul Kim, Jingxu Zhang, Wonjoon Jin, Sunghyun Cho, Qi Dai, **Jaesik Park**, and Chong Luo
Learning Zero-Shot Subject-Driven Video Generation Using 1% Compute
European Conf. on Computer Vision (**ECCV**), 2026

- [5] Ruihan Gao, Joonghyuk Shin, Ava Pun, **Jaesik Park**, Wenzhen Yuan, and Jun-Yan Zhu
Text-based Tactile Graphics Generation for the Visually Impaired
European Conf. on Computer Vision (**ECCV**), 2026
- [6] Suwoong Yeom, Joonsik Nam, Seunggyu Choi, Lucas Yunkyu Lee, Sangmin Kim, **Jaesik Park**, Joonsoo Kim, Kugjin Yun, Kyeongbo Kong, and Sukju Kang
TRiGS: Temporal Rigid-Body Motion for Scalable 4D Gaussian Splatting
European Conf. on Computer Vision (**ECCV**), 2026
- [7] Sanghyun Jo, Wooyeol Lee, Ziseok Lee, Jonghyun Choi, **Jaesik Park**, and Kyungsu Kim
ISAC: Training-Free Instance-to-Semantic Attention Control for Multi-Instance Generation
European Conf. on Computer Vision (**ECCV**), 2026
- [8] Mingi Kwon*, Joonghyuk Shin*, Jaeseok Jung, **Jaesik Park**†, and Youngjung Uh†
JAM-Flow: Joint Audio-Motion Synthesis with Flow Matching
European Conf. on Computer Vision (**ECCV**), 2026
(*Equal Contribution, †Equal Advising)
- [9] Minhyuk Hwang*, Sangmin Kim*, Seunguk Do, Daneul Kim, and **Jaesik Park**
4D Human-Scene Reconstruction from Low-Overlap Captures
ACM Special Interest Group on Graphics and Interactive Techniques (**SIGGRAPH**), 2026
(*Equal contribution)
- [10] Lucas Yunkyu Lee, Soonho Kim, Youngwook Kim, Sangmin Kim, and **Jaesik Park**
FreeTimeGS++: Secrets of Dynamic Gaussian Splatting and Their Principles
2605.03337 (arXiv), 2026
- [11] Sangmin Kim, Minhyuk Hwang, Geonho Cha, Dongyoon Wee, and **Jaesik Park**
Coherent Human-Scene Reconstruction from Multi-Person Multi-View Video in a Single Pass
2603.12789 (arXiv), 2026
- [12] Seungwoo Yoon, Jinmo Kim, and **Jaesik Park**
Extend3D: Town-scale 3D Generation
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2026
- [13] Pierre Musacchio, Jaeyi Jeong, Dahun Kim, and **Jaesik Park**
Taking Shortcuts for Categorical VQA using Super Neurons
2603.10781 (arXiv), 2026
- [14] Youngjoong Kim, Duhoe Kim, Woosung Kim, and **Jaesik Park**
Stabilizing Consistency Training: A Flow Map Analysis and Self-Distillation
2601.22679 (arXiv), 2026
- [15] Seunguk Do, Minwoo Huh, Joonghyuk Shin, and **Jaesik Park**
Direct Reward Fine-Tuning on Poses for Single Image to 3D Human in the Wild
Int. Conf. on Learning Representations (**ICLR**), 2026
- [16] Sanghyun Jo*, Ziseok Lee*, Wooyeol Lee, Jonghyun Choi, **Jaesik Park**†, and Kyungsu Kim†
TRACE: Your Diffusion Model is Secretly an Instance Edge Detector
Int. Conf. on Learning Representations (**ICLR**), 2026
(*Equal contribution, †Corresponding authors, **Oral** presentation, 1.13% acceptance rate)
- [17] Joonghyuk Shin, Zhengqi Li, Richard Zhang, Jun-Yan Zhu, **Jaesik Park**, Eli Shechtman, and Xun Huang
MotionStream: Real-Time Video Generation with Interactive Motion Controls
Int. Conf. on Learning Representations (**ICLR**), 2026
(**Oral** presentation, 1.13% acceptance rate)
- [18] Haebeom Jung, Nantae Kim, Jungwoo Kim, and **Jaesik Park**
Targetless LiDAR-Camera Calibration with Neural Gaussian Splatting
IEEE Robotics and Automation Letters (**RAL**), 2026
(Invited to present at **ICRA 2026**)

- [19] Hyungtae Lim*, Minkyun Seo*, Luca Carlone, and **Jaesik Park**
Towards Zero-Shot Point Cloud Registration Across Diverse Scales, Scenes, and Sensor Setups
2601.02759 (arXiv), 2026
(*Equal contribution)
- [20] Deepak Ghimire, Dayoung Kil, Seonghwan Jeong, **Jaesik Park**, and Seong-Heum Kim
One-Cycle Structured Pruning via Stability-Driven Subnetwork Search
Winter Conf. on Applications of Computer Vision (**WACV**), 2026
- [21] Kanghee Lee, Injae Lee, Minseok Kwak, Kwonyoung Ryu, Jungi Hong, and **Jaesik Park**
SpatialMosaic: A Multiview VLM Dataset for Partial Visibility
2512.23365 (arXiv), 2025
- [22] Seunghun Oh, Jaesung Choe, Dongjae Lee, Daeun Lee, Seunghoon Jeong, Yu-Chiang Frank Wang, and **Jaesik Park**
SVRecon: Sparse Voxel Rasterization for Surface Reconstruction
2511.17364 (arXiv), 2025
- [23] Minseo Lee, Byeonghyeon Lee, Lucas Yunkyu Lee, Eunsoo Lee, Sangmin Kim, Seunghyeon Song, Joo Chan Lee, Jong Hwan Ko, **Jaesik Park**, and Eunbyung Park
Optimized Minimal 4D Gaussian Splatting
2510.03857 (arXiv), 2025
- [24] In-Jae Lee, Mungyeom Kim, Kwonyoung Ryu, Pierre Musacchio, and **Jaesik Park**
OpenBox: Annotate Any Bounding Boxes in 3D
Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2025
(Accepted as a **spotlight** paper)
- [25] Hyeonseong Jeon, Cheolhong Min, and **Jaesik Park**
Tree-Guided Diffusion Planner
Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2025
- [26] Pierre Musacchio, Hyunmin Lee, and **Jaesik Park**
Holistic Order Prediction in Natural Scenes
Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2025
- [27] Hyunjin Kim, Haebeom Jung, and **Jaesik Park**
Metropolis-Hastings Sampling for 3D Gaussian Reconstruction
Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2025
- [28] Minkyun Seo*, Hyungtae Lim*, Kanghee Lee, Luca Carlone, and **Jaesik Park**
BUFFER-X: Towards Zero-Shot Point Cloud Registration in Diverse Scenes
Int. Conf. on Computer Vision (**ICCV**), 2025
(Accepted as a **highlight paper**, *Joint first authors)
- [29] Hyunjoon Lee, Joonkyu Min, and **Jaesik Park**
CF3: Compact and Fast 3D Feature Fields
Int. Conf. on Computer Vision (**ICCV**), 2025
(Also accepted at **ICCV 2025 demonstrations track**)
- [30] Joonghyuk Shin, Alchan Hwang, Yujin Kim, Daneul Kim, and **Jaesik Park**
Exploring Multimodal Diffusion Transformers for Enhanced Prompt-based Image Editing
Int. Conf. on Computer Vision (**ICCV**), 2025
- [31] Seungjoo Shin, **Jaesik Park**, and Sunghyun Cho
Leveraging Learned Image Prior for 3D Gaussian Compression
Workshop on Efficient Computing under Limited Resources (ECLR) **ICCV workshop**, 2025
- [32] Sangmin Kim, Seunguk Do, and **Jaesik Park**
ShowMak3r: Compositional TV Show Reconstruction
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2025
- [33] Jaeah Lee, Changwoon Choi, Young Min Kim, and **Jaesik Park**
Recovering Dynamic 3D Sketches from Videos
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2025

- [34] Daneul Kim, Jaeah Lee, and **Jaesik Park**
Improving Editability in Image Generation with Layer-wise Memory
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2025
- [35] Hyungtae Lim, Daebeom Kim, Gunhee Shin, Jingnan Shi, Ignacio Vizzo, Hyun Myung, **Jaesik Park**, and Luca Carlone
KISS-Matcher: Fast and Robust Point Cloud Registration Revisited
Int. Conf. on Robotics and Automation (**ICRA**), 2025
- [36] Seungjoo Shin, **Jaesik Park**, and Sunghyun Cho
Locality-aware Gaussian Compression for Fast and High-quality Rendering
Int. Conf. on Learning Representations (**ICLR**), 2025
- [37] Ashish Kumar and **Jaesik Park**
Designing Concise ConvNets with Columnar Stages
Int. Conf. on Learning Representations (**ICLR**), 2025
- [38] Ashish Kumar and **Jaesik Park**
Cross Resolution Encoding-Decoding For Detection Transformers
2410.04088 (arXiv), 2024
- [39] Seokjun Ahn*, Jungtaek Kim*, Minsu Cho, and **Jaesik Park**
Budget-Aware Sequential Brick Assembly with Efficient Constraint Satisfaction
Transactions on Machine Learning Research (**TMLR**), 2024
(*Joint first authors)
- [40] Joonghyuk Shin, Daehyeon Choi, and **Jaesik Park**
InstantDrag: Improving Interactivity in Drag-based Image Editing
ACM Special Interest Group on Graphics and Interactive Techniques (**SIGGRAPH Asia**), 2024
- [41] Minguk Kang, Richard Zhang, Connelly Barnes, Sylvain Paris, Suha Kwak, **Jaesik Park**, Eli Shechtman, Jun-Yan Zhu, and Taesung Park
Distilling Diffusion Models into Conditional GANs
European Conf. on Computer Vision (**ECCV**), 2024
- [42] Jungeon Kim, Soongjin Kim, **Jaesik Park**, and Seungyong Lee
Deep Cost Ray Fusion for Sparse Depth Video Completion
European Conf. on Computer Vision (**ECCV**), 2024
- [43] Nahyuk Lee, Juhong Min, Junha Lee, Seungwook Kim, Kanghee Lee, **Jaesik Park**, and Minsu Cho
3D Geometric Shape Assembly via Efficient Point Cloud Matching
Int. Conf. on Machine Learning (**ICML**), 2024
- [44] Changwoon Choi, Jaeah Lee, **Jaesik Park**, and Young Min Kim
3Doodle: Compact Abstraction of Objects with 3D Strokes
ACM Special Interest Group on Graphics and Interactive Techniques (**SIGGRAPH**), 2024
(Accepted as a journal track paper)
- [45] Seoyeon Kim, Minguk Kang, Dongwon Kim, **Jaesik Park**, and Suha Kwak
Extending CLIP’s Image-Text Alignment to Referring Image Segmentation
Annual Conf. of the North American Chapter of the Assoc. for Computational Linguistics (**NAACL**), 2024
- [46] Ashish Kumar, Daneul Kim, **Jaesik Park**, and Laxmidhar Behera
Pick-or-Mix: Dynamic Channel Sampling for ConvNets
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2024
- [47] Chunghyun Park, Seungwook Kim, **Jaesik Park**, and Minsu Cho
Learning $SO(3)$ -Invariant Semantic Correspondence via Local Shape Transform
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2024
- [48] Ashish Kumar, **Jaesik Park**, and Laxmidhar Behera
High-Speed Stereo Visual SLAM for Low-Powered Computing Devices
IEEE Robotics and Automation Letters (**RAL**), vol. 9, issue 1, 2024
(Invited for an **oral** presentation at **ICRA 2024**)

- [49] Seungjoo Shin and **Jaesik Park**
Binary Radiance Fields
 Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2023
 (Received 30th **HumanTech Paper Award (Silver Prize)**, Samsung Electronics Corp.)
- [50] MinGuk Kang, Joonghyuk Shin, and **Jaesik Park**
StudioGAN: A Taxonomy and Benchmark of GANs for Image Synthesis
 Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2023
- [51] Jaesung Choe, Christopher Choy, **Jaesik Park**, In So Kweon, and Animashree Anandkumar
Spacetime Surface Regularization for Neural Dynamic Scene Reconstruction
 Int. Conf. on Computer Vision (**ICCV**), 2023
- [52] Joonghyuk Shin, Minguk Kang, and **Jaesik Park**
Fill-Up: Balancing Long-Tailed Data with Generative Models
 2306.07200 (arXiv), 2023
- [53] Seungwook Kim, Chunghyun Park, Yoonwoo Jeong, **Jaesik Park**, and Minsu Cho
Stable and Consistent Prediction of 3D Characteristic Orientation via Invariant Residual Learning
 Int. Conf. on Machine Learning (**ICML**), 2023
- [54] Minguk Kang, Jun-Yan Zhu, Richard Zhang, **Jaesik Park**, Eli Shechtman, Sylvain Paris, and Taesung Park
Scaling up GANs for Text-to-Image Synthesis
 Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2023
 (Accepted as a **highlight paper** (review score 5,5,5) - top 2.5% among 9,155 submissions)
- [55] Kwonyoung Ryu, Soonmin Hwang, and **Jaesik Park**
Instant Domain Augmentation for LiDAR Semantic Segmentation
 Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2023
- [56] Kanghee Lee, Junha Lee, and **Jaesik Park**
Learning to Register Unbalanced Point Pairs
 3D Vision and Robotics **CVPR workshop**, 2023
- [57] Jinoh Cho, Minguk Kang, Vibhav Vineet, and **Jaesik Park**
Instance-Aware Image Completion
 AI for Content Creation (AI4CC) **CVPR workshop**, 2023
- [58] Rongrong Gao, Tian-Zhu Xiang, Chenyang Lei, **Jaesik Park**, and Qifeng Chen
Scene-level Point Cloud Colorization with Semantic-and-Geometric-aware Networks
 IEEE Int. Conf. on Robotics and Automation (**ICRA**), 2023
- [59] Seungjoo Shin*, Min Woo Kim*, Kyong Hwan Jin, Kwang Moo Yi, Yoshiki Kohmura, Tetsuya Ishikawa, Jung Ho Je, and **Jaesik Park**
Deep 3D Reconstruction of Synchrotron X-ray Computed Tomography for Intact Lungs
 published by Nature Research (**Scientific Reports**), 2023
 (*Equal contribution)
- [60] Seokjun Ahn, Jungtaek Kim, Minsu Cho, and **Jaesik Park**
Sequential Brick Assembly with Efficient Constraint Satisfaction
 2210.01021 (arXiv), 2022
- [61] Hyomin Kim, Hyeonseo Nam, Jungeon Kim, **Jaesik Park**, and Seungyong Lee
LaplacianFusion: Detailed 3D Clothed-Human Body Reconstruction
 ACM Special Interest Group on Graphics and Interactive Techniques (**SIGGRAPH Asia**), 2022
 (Accepted as a journal track paper)
- [62] Yoonwoo Jeong*, Seungjoo Shin*, Junha Lee*, Christopher Choy, Animashree Anandkumar, Minsu Cho, and **Jaesik Park**
PeRFception: Perception using Radiance Fields
 Int. Conf. on Neural Information Processing Systems (**NeurIPS**) Datasets and Benchmarks Track, 2022
 (*Equal contribution)

- [63] Nayeong Kim, Sehyun Hwang, Sungsoo Ahn, **Jaesik Park**, and Suha Kwak
Learning Debiased Classifier with Biased Committee
Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2022
- [64] Seunghyuk Cho, Juyong Lee, **Jaesik Park**, and Dongwoo Kim
A Rotated Hyperbolic Wrapped Normal Distribution for Hierarchical Representation Learning
Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2022
- [65] Seungwook Kim, Yoonwoo Jeong, Chunghyun Park, **Jaesik Park**, and Minsu Cho
SeLCA: Self-Supervised Learning of Canonical Axis
Symmetry and Geometry in Neural Representations (NeurReps), **NeurIPS workshop**, 2022
- [66] Jiye Kim, Seungbeom Lee, Dongwoo Kim, Sungsoo Ahn, and **Jaesik Park**
Substructure-Atom Cross Attention for Molecular Representation Learning
AI for Science: Progress and Promises (AI4Science) **NeurIPS workshop**, 2022
- [67] Juyong Lee*, Seokjun Ahn*, and **Jaesik Park**
Style-Agnostic Reinforcement Learning
European Conf. on Computer Vision (**ECCV**), 2022
(*Equal contribution)
- [68] Jaesung Choe*, Chunghyun Park*, Francois Rameau, **Jaesik Park**, and In So Kweon
PointMixer: MLP-Mixer for Point Cloud Understanding
European Conf. on Computer Vision (**ECCV**), 2022
(*Equal contribution)
- [69] Jaewon Kam, Jungeon Kim, Soongjin Kim, **Jaesik Park**, and Seungyong Lee
CostDCNet: Cost Volume based Depth Completion for a Single RGB-D Image
European Conf. on Computer Vision (**ECCV**), 2022
- [70] Jinhwi Lee, Jungtaek Kim, Hyunsoo Chung, **Jaesik Park**, and Minsu Cho
Learning to Assemble Geometric Shapes
Int. Joint Conf. on Artificial Intelligence (**IJCAI**), 2022
- [71] Hyunmin Lee and **Jaesik Park**
Instance-wise Occlusion and Depth Orders in Natural Scenes
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2022
- [72] Chunghyun Park, Yoonwoo Jeong, Minsu Cho, and **Jaesik Park**
Fast Point Transformer
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2022
(Received 28th **HumanTech Paper Award (Silver Prize)**, Samsung Electronics Corp.)
- [73] Jaebong Jeong, Janghun Jo, Sunghyun Cho, and **Jaesik Park**
3D Scene Painting via Semantic Image Synthesis
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2022
- [74] Jungeon Kim, Hyomin Kim, Hyeonseo Nam, **Jaesik Park**, and Seungyong Lee
TextureMe: High-quality Textured Scene Reconstruction in Real-time
ACM Transactions on Graphics (**ToG**), 2022
(also presented at **SIGGRAPH2022**)
- [75] Jaesung Choe, Byeongin Joung, Francois Rameau, **Jaesik Park**, and In So Kweon
Deep Point Cloud Reconstruction
Int. Conf. on Learning Representations (**ICLR**), 2022
- [76] Jae Shin Yoon, Zhixuan Yu, **Jaesik Park**, and Hyun Soo Park
HUMBI: A Large Multiview Dataset of Human Body Expressions and Benchmark Challenge
Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2021
- [77] Junha Lee, Christopher Choy, and **Jaesik Park**
Putting 3D Spatially Sparse Networks on a Diet
2112.01316 (arXiv), 2021

- [78] Wei Dong*, Kwonyoung Ryu*, Michael Kaess, and **Jaesik Park**
Revisiting LiDAR Registration and Reconstruction: A Range Image Perspective
 2112.02779 (arXiv), 2021
 (*Equal contribution)
- [79] Jinsoo Choi, **Jaesik Park**, and In So Kweon
Self-Supervised Real-time Video Stabilization
 British Machine Vision Conference (**BMVC**), 2021
- [80] Minguk Kang, Woohyeon Shim, Minsu Cho, and **Jaesik Park**
Rebooting ACGAN: Auxiliary Classifier GANs with Stable Training
 Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2021
- [81] Hyunsoo Chung, Jungtaek Kim, Boris Knyazev, Jinhwi Lee, Graham W. Taylor, **Jaesik Park**, and Minsu Cho
Brick-by-Brick: Combinatorial Construction with Deep Reinforcement Learning
 Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2021
- [82] Yoonwoo Jeong, Seokjun Ahn, Christopher Choy, Animashree Anandkumar, Minsu Cho, and **Jaesik Park**
Self-Calibrating Neural Radiance Fields
 Int. Conf. on Computer Vision (**ICCV**), 2021
- [83] Junha Lee, Seungwook Kim, Minsu Cho, and **Jaesik Park**
Deep Hough Voting for Robust Global Registration
 Int. Conf. on Computer Vision (**ICCV**), 2021
- [84] Hyomin Kim, Jungeon Kim, Jaewon Kam, **Jaesik Park***, and Seungyong Lee*
Deep Virtual Markers for Articulated 3D Shapes
 Int. Conf. on Computer Vision (**ICCV**), 2021
 (*Joint corresponding authors, **Oral Presentation**, 3.4% acceptance rate)
- [85] Hyunmin Lee and **Jaesik Park**
STAD: Stable Video Depth Estimation
 IEEE Int. Conf. on Image Processing (**ICIP**), 2021
- [86] Taewon Jin, Taesoo Park, Ina Park, **Jaesik Park***, and Ji Hoon Shim*
Accelerated Crystal Structure Prediction of Multi-elements Random Alloy using Expandable Features
 published by Nature Research (**Scientific Reports**), 2021
 (*Joint corresponding authors)
- [87] Hyomin Kim, Jungeon Kim, Hyeonseo Nam, **Jaesik Park**, and Seungyong Lee
Spatiotemporal Texture Reconstruction for Dynamic Objects Using a Single RGB-D Camera
 42nd Annual Conference of the European Association for Computer Graphics (**EuroGraphics**), 2021
- [88] Jinsoo Choi, **Jaesik Park**, and In So Kweon
High-quality Frame Interpolation via Tridirectional Inference
 Winter Conf. on Applications of Computer Vision (**WACV**), 2021
- [89] Minguk Kang and **Jaesik Park**
ContraGAN: Contrastive Learning for Conditional Image Generation
 Int. Conf. on Neural Information Processing Systems (**NeurIPS**), 2020
- [90] Jungtaek Kim, Hyunsoo Chung, Minsu Cho, and **Jaesik Park**
Combinatorial 3D Shape Generation via Sequential Assembly
 Machine Learning for Engineering Modeling, Simulation, and Design (ML4Eng), **NeurIPS workshop**, 2020
- [91] Jinhwi Lee*, Jungtaek Kim*, Hyunsoo Chung, **Jaesik Park**, and Minsu Cho
Fragment Relation Networks for Geometric Shape Assembly
 Learning Meets Combinatorial Algorithms (LMCA), **NeurIPS workshop**, 2020
 (*Equal contribution)
- [92] Zhixuan Yu, Jaeshin Yoon, Inkyu Lee, Prashanth Venkatesh, **Jaesik Park**, Jihun Yu, and Hyunsoo Park
HUMBI: A Large Multiview Dataset of Human Body Expressions
 Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2020

- [93] Christopher Choy, Junha Lee, Rene Ranftl, **Jaesik Park**, and Vladlen Koltun
High-Dimensional Convolutional Networks for Geometric Pattern Recognition
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2020
(**Oral** Presentation, 5.7% acceptance rate)
- [94] Yue Wu, Rongrong Gao, **Jaesik Park**, and Qifeng Chen
Future Video Synthesis with Object Motion Predictions
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2020
- [95] Christopher Choy*, **Jaesik Park***, and Vladlen Koltun
Fully Convolutional Geometric Features
Int. Conf. on Computer Vision (**ICCV**), 2019
(*Equal contribution)
- [96] Jungeon Kim, Hyomin Kim, **Jaesik Park**, and Seungyong Lee
Global Texture Mapping for Dynamic Objects
Pacific Graphics (**PG**), 2019
- [97] Wei Dong, **Jaesik Park**, Yi Yang, and Michael Kaess
GPU Accelerated Robust Scene Reconstruction
IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (**IROS**), 2019
- [98] Hae-Gon Jeon, **Jaesik Park**, Gyeongmin Choe, Jinsun Park, Yunsu Bok, Yu-Wing Tai, and In So Kweon
Depth from a Light Field Image with Learning-based Matching Costs
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2019
- [99] Maxim Tatarchenko*, **Jaesik Park***, Vladlen Koltun, and Qian-Yi Zhou
Tangent Convolutions for Dense Prediction in 3D
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2018
(*Equal contribution. **Spotlight Oral** Presentation)
- [100] Qian-Yi Zhou, **Jaesik Park**, and Vladlen Koltun
Open3D: A Modern Library For 3D Data Processing
1801.09847 (arXiv), 2018
- [101] Byungtae Ahn, Dong-Geol Choi, **Jaesik Park**, and In So Kweon
Real-time Head Pose Estimation using Multi-task Deep Neural Network
Robotics and Autonomous Systems (**RAS**), 2018
- [102] **Jaesik Park**, Qian-Yi Zhou, and Vladlen Koltun
Colored Point Cloud Registration Revisited
Int. Conf. on Computer Vision (**ICCV**), 2017
- [103] Arno Knapitsch, **Jaesik Park**, Qian-Yi Zhou, and Vladlen Koltun
Tanks and Temples: Benchmarking Large-Scale Scene Reconstruction
ACM Special Interest Group on Graphics and Interactive Techniques (**SIGGRAPH**), 2017
- [104] Gyeongmin Choe, **Jaesik Park**, Yu-Wing Tai, and In So Kweon
Refining Geometry from Depth Sensors using IR Shading Images
International Journal of Computer Vision (**IJCV**), 2017
- [105] Seong heum Kim, Yu Wing Tai, Joon Young Lee, **Jaesik Park**, and In So Kweon
Category Specific Salient View Selection via Deep Convolutional Neural Networks
Computer Graphics Forum (**CGF**), 2017
- [106] **Jaesik Park**, Sudipta N. Sinha, Yasuyuki Matsushita, Yu-Wing Tai, and In So Kweon
Robust Multiview Photometric Stereo using Planar Mesh Parameterization
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2016
- [107] Qian-Yi Zhou, **Jaesik Park**, and Vladlen Koltun
Fast Global Registration
European Conf. on Computer Vision (**ECCV**), 2016
(**Oral** Presentation, 1.8% acceptance rate)

- [108] **Jaesik Park**, Yu-Wing Tai, Sudipta N. Sinha, and In So Kweon
Efficient and Robust Color Consistency for Community Photo Collections
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2016
- [109] Hyowon Ha, Sunghoon Im, **Jaesik Park**, Hae-Gon Jeon, and In So Kweon
High-quality Depth from Uncalibrated Small Motion Clip
Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2016
(**Oral** Presentation, 3.9% acceptance rate)
- [110] Inwook Shim, Seunghak Shin, Yunsu Bok, Kyungdon Joo, Dong-Geol Choi, Joon-Young Lee, **Jaesik Park**, Jun Ho Oh, and In So Kweon
Vision System and Depth Processing for DRC-HUBO+
IEEE Int. Conf. on Robotics and Automation (**ICRA**), 2016
(Depth processing algorithm of Team KAIST (winner of DARPA robotics challenge finals 2015))
- [111] Seong-Heum Kim, Yu-Wing Tai, **Jaesik Park**, and In So Kweon
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PROGRAM COMMITTEE

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- Area Chair, Machine Vision Applications (MVA) conference, 2021
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- Senior Program Committee, Int. Joint Conf. on Artificial Intelligence (**IJCAI**), 2021
- Senior Program Committee, Assoc. for the Advancement of Artificial Intelligence (**AAAI**), 2021
- **Area Chair**, Int. Conf. on Computer Vision and Pattern Recognition (**CVPR**), 2020
- Senior Program Committee, Int. Joint Conf. on Artificial Intelligence (**IJCAI**), 2020
- **Area Chair**, Int. Conf. on Computer Vision (**ICCV**), 2019
- **Session Chair**, Int. Conf. on Computer Vision (**ICCV**), 2019
- Have been served as a reviewer for international conferences, such as **CVPR**, **ICCV**, **ECCV**, **ICLR**, **NeurIPS**, **AAAI**, **ICRA**, **IROS**, **SIGGRAPH**, **SIGGRAPH Asia**, **BMVC**, **3DV**, **ACCV**, **WACV**, and so on.
- Have been served as a reviewer for international journals, such as **TPAMI**, **TIP**, **TVCG**, **TRO**, **IJCV**, **CVIU**, **SPL**, **IVC**, **Neurocomputing**, and so on.

- **Academic Grant Program**, NVIDIA Corp., Sep. 2025
- **Gift Grant**, Microsoft Corp., Sep. 2024
- **30th HumanTech Paper Award (Silver Prize)**, Samsung Electronics Corp., Feb. 2024
- **Representative ICT RnD Projects** in Recent Five Years (ICT RnD 사업 성과 중 우수과제), IITP (정보통신기획평가원), Republic of Korea, May 2022
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- **Faculty Support Program**, Intel Corporation, June 2020
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- AIGS101, Artificial Intelligence Basis II, Fall, 2020
- CSED100, Introduction to Computer Science Engineering, Fall, 2020
- CSED703F, 3D Vision, Fall, 2020

- CSED800/AIGS800, CSE/GSAI Colloquium, Fall, 2020
- AIGS101, Artificial Intelligence Basis I, Spring, 2020
- AIGS537/CSED537, Artificial Intelligence and Data Science Spring, Spring, 2020
- CSED233, Data Structure, **Awarded as the Outstanding Online Class** by POSTECH, Spring, 2020
- CSED703F-01 3D Vision, Fall, 2019

FUNDING

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- 3D Point Cloud 기반의 환경 인식 기술 개발, LG Electronics, 2022.06-2022.12
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- *Discovering Visual Common Sense*, Qualcomm, USA, 2022.03-2023.03
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- 문화체육관광부, 포스텍-한예중 문화기술선도대학원, Ministry of Culture, Sports and Tourism, Republic of Korea, 2021.06-2023.12
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- Realistic 3D Reconsturction and Rendering for Real-World Simulations, 한국자동차공학회 융합S/W 및 AI부분 워크숍, Oct. 2025
- Realistic 3D Reconsturction and Rendering for Real-World Simulations, 서울대학교 2기 빅데이터 AI CEO과정, Oct. 2025
- Holistic Order Prediction in Natural Scenes, Global AI Frontier Lab, Oct. 2025
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- 3D Vision Research from SNU VGI Lab, **Naver Labs**, Seongnam, Republic of Korea, March 2025
- 공간표현과 공간지능 최신동향, 과실연 AI 미래 포럼, Youtube Live, March 2025
- 영상기반 3차원 복원과 AI 응용, 한국인공지능학회 인공지능 동계 단기강좌, Seoul, Republic of Korea, Feb. 2025
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- Social Speed Mentoring, **ECCV 2024**, Milano, Italy, Sep. 2024
- Radiance Field Reconstruction, **KIA** motor company, Gwangmyeong, Republic of Korea, Aug. 2024
- Radiance Field Reconstruction, Koh Young Technology, Yongin, Republic of Korea, Aug. 2024
- 가우시안 스피래팅 기법의 발전 현황, 공간컴퓨팅을 위한 지능형 XR 기술 및 응용 워크숍, Seoul, RoK, July 2024
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- 3D Scene Understanding, 삼성전자 DS사업부 특강, Seoul, Republic of Korea, June 2024
- 3D Scene Reconstruction, Representation, and Understanding, **KAIST** 초세대연구실 개소 기념 특별 심포지엄, Daejeon, Republic of Korea, Feb. 2024
- Topics in 3D vision integrating AI modules, **Hyundai Motor Group**, Virtual, Oct. 2023
- Learning to Understand 3D Point Clouds, 2023 Fall Workshop on Algorithms and Computation, **POSTECH**, Pohang, Republic of Korea, Oct. 2023
- Towards NeRF at Scale, Computing Frontier Summer School, **Seoul Nat. Univ.**, Republic of Korea, Aug. 2023
- Storage Efficient Radiance Fields, **ETRI**, Daejeon, Republic of Korea, July 2023
- Social Speed Mentoring, **CVPR 2023**, Vancouver, Canada, June 2023
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- 전자공학회 영상이해 연구회 겨울학교, 대한전자공학회, Hoengseong, Republic of Korea, Jan. 2023
- 3D Vison and Open3D, **Seoul National University**, Seoul, Republic of Korea (offline+online), Oct. 2022

- *Object Detection from Images or Point Clouds*, **LG Electronics**, Seoul (online), Republic of Korea, Aug. 2022
- *Fast Point Transformer*, **Harvard University**, Boston, MA, USA (online), June 2022
- *Self-Calibrating Neural Radiance Fields*, **Seoul National University**, Seoul, Republic of Korea (online), Feb. 2022
- *Self-Calibrating Neural Radiance Fields*, **DGIST**, Daegu, Republic of Korea (online), Aug. 2021
- *Object Detection from Images or Point Clouds*, **LG Electronics**, Seoul, Republic of Korea, Aug. 2021
- *Self-Calibrating Neural Radiance Fields*, Korea Institute of Science and Technology (**KIST**), Virtual, June 2021
- *Mentor Session: How to Become a Professor*, **CVPR 2021**, Virtual, June 2021
- *Recent Work on Image Generation*, **GIST**, Gwangju, Republic of Korea (online), May 2021
- *Point Cloud Registration using Hierarchical Hough Transform*, 33rd Workshop on Image Processing and Image Understanding (**IPIU 2021**), Virtual, Feb. 2021
- *3D Representations and Detections*, Chungbuk University, Virtual, Jan. 2021
- *Introduction to Computer Vision*, **Kyungbuk Science High School**, Pohang, Republic of Korea, Aug. 2020
- *High-Dimensional Convolutional Networks for Geometric Pattern Recognition*, **KCCV 2020**, Seoul, Aug. 2020
- *Object Detection from Images or Point Clouds*, **LG Electronics**, Seoul, Republic of Korea, Aug. 2020
- *Geometric Pattern Recognition*, 32nd Workshop on Image Proc. and Image Understanding (**IPIU 2020**), Feb. 2020
- *Fully Convolutional Geometric Features*, Koh Young Technology, Yongin, Republic of Korea, Dec. 2019
- *Open3D Tutorial and Fully Convolutional Geometric Features*, **KETI**, Seongnam, Republic of Korea, Nov. 2019
- *Introduction to Computer Vision and Deep Learning*, **Daegu Science High School**, Aug. 2019
- *3D Computer Vision and Open3D*, Int. Conf. on Machine Vision Applications (**MVA**), Tokyo, Japan, May 2019
- *3D Computer Vision and Open3D*, **Qualcomm**, San Diego, USA, June 2019
- *3D reconstruction using Open3D*, Minneapolis, **University of Minnesota**, invited lecture for Multiview 3D Geometry in Computer Vision (CSCI 5980) Class, Apr. 2018
- *3D reconstruction using Open3D*, **Forma Technology** (acquired by **Snap Inc.**), San Francisco, USA, March 2018

OTHER WORKING EXPERIENCES

Post-Doc. Researcher	Daejeon, Republic of Korea
KAIST (<i>Mentor: Prof. In So Kweon</i>)	Aug. 2015 – Nov. 2015
Research Intern	Redmond, WA, USA
<i>Microsoft Research (MSR)</i> (<i>Mentor: Dr. Sudipta N. Sinha</i>)	June 2013 – Sep. 2013
Research Intern	Beijing, China
<i>Microsoft Research Asia (MSRA)</i> (<i>Mentor: Prof. Y. Matsushita</i>)	April 2012 – Oct. 2012

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